Integer quadruples and primitive three-square representations

Product formulas, equality cases, and density zero for A354766/A278085

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Draft for independent human mathematical review

Status and scope. This is a conventional mathematical proof, with its standard inputs stated explicitly. A companion Lean development also verifies the representation formula and the arithmetic, equality, and density conclusions without proof admissions. Its final dependency audits use only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. Independent external human mathematical review is pending.

Abstract

We count ordered integer quadruples whose sum is a positive integer n and whose sum of squares is n^2. A Hadamard transformation gives a precise primitive-counting bridge, including the different normalizations for odd n and for $n \equiv 2 \pmod 4$. We prove the needed primitive three-square product by an explicit lattice-neighbor double count. The only non-elementary standard inputs are Minkowski's convex-body theorem and quadratic reciprocity; no three-square counting formula is assumed. The resulting normalized primitive and total counts are multiplicative, with explicit prime-power values. We characterize exactly when the total count equals n and prove that this equality set has natural density zero by a finite arithmetic sieve. The sieve requires no theorem about reciprocal primes in an arithmetic progression.

1 Definitions and proof dependencies

For $n \geq 1$, let $\mathcal{Q}_n$ be the set of ordered quadruples $(h, i, j, k) \in \mathbb{Z}^4$ satisfying

$$h + i + j + k = n, \qquad h^2 + i^2 + j^2 + k^2 = n^2. \tag{1}$$

Signs and coordinate orders are counted separately. Define the content of a quadruple to be $\gcd(|h|, |i|, |j|, |k|)$; a quadruple is *primitive* when its content is one. Put

$$\mathrm{tq}(n) = |\mathcal{Q}_n|, \qquad \mathrm{pq}(n) = |\{q \in \mathcal{Q}_n : q \text{ is primitive}\}|.$$

These sets are finite: every coordinate has absolute value at most n. Following the normalization of the OEIS pair [1, 2], put initially as rational numbers

$$T(n) = \frac{\mathrm{tq}(n)}{4}, \qquad P(n) = \frac{\mathrm{pq}(n)}{4}. \tag{2}$$

We will prove divisibility of both raw counts by four before using these as integer-valued arithmetic functions.

For a positive integer M, let $r_3^{\mathrm{prim}}(M)$ count ordered integer triples (x, y, z) with $x^2 + y^2 + z^2 = M$ and $\gcd(|x|, |y|, |z|) = 1$. In particular, define

$$F(m) = r_3^{\mathrm{prim}}(3m^2) \qquad (m \geq 1).$$

All these sets are finite by the same coordinate bound. For an odd prime p, let $\chi_p = \left(\frac{-3}{p}\right)$ be the Legendre symbol, with value zero when $p = 3$.

The lattice construction below is the classical neighbor construction of Kneser [5]; Voight [6], Proposition 3.13 and Corollary 3.15, provide background. We prove the particular constructions, low-rank classification, and representation double count needed here. Cooper–Hirschhorn [7] give an independent published cross-check of the product formula, but their representation theorem is not an input to this proof.

The standard inputs used without reproof are Minkowski's theorem in the following form: a centrally symmetric convex open set in $\mathbb{R}^r$ of volume strictly greater than 2^r times a lattice's covolume contains a nonzero lattice vector [3]; and quadratic reciprocity with the supplementary law for -1 [4]. We also use elementary integer factorization, the Chinese remainder theorem, finite-dimensional linear algebra, and finite sums and products. No additional representation formula or analytic assertion about primes is assumed.

2 The corrected Hadamard counting bridge

For a quadruple satisfying (1), put

$$x = h + i - j - k, \quad y = h - i + j - k, \quad z = h - i - j + k.$$

Expansion gives

$$x^2 + y^2 + z^2 + (h + i + j + k)^2 = 4(h^2 + i^2 + j^2 + k^2),$$

so $x^2 + y^2 + z^2 = 3n^2$. Conversely, the inverse expressions are

$$\begin{aligned} 4h &= n + x + y + z, \quad 4i = n + x - y - z, \\ 4j &= n - x + y - z, \quad 4k = n - x - y + z. \end{aligned} \tag{3}$$

Whenever their right sides are divisible by four, these expressions give an integer quadruple satisfying (1). Indeed, their sum is $4n$; the same Hadamard norm identity gives the sum of squares n^2. Thus these linear maps are inverse bijections after the integrality condition is imposed.

Lemma 2.1 (A common divisor)**.** *If d, m are positive integers and $d^2 \mid 3m^2$, then $d \mid m$.*

Proof. For every prime p, write $a = v_p(d)$ and $b = v_p(m)$. If $p \neq 3$, the divisibility gives $2a \leq 2b$. For $p = 3$ it gives $2a \leq 1 + 2b$, which also implies $a \leq b$ because both are integers. This proves the assertion prime by prime. □

The primary definition of A278085 uses primitive quadruples of sum $3n$ and squared norm $3n^2$. Its equivalence with $P(n)$ is explicit: the map

$$(h, i, j, k) \longmapsto (n - h, n - i, n - j, n - k)$$

is an involution interchanging these quadruples with those in $\mathcal{Q}_n$. Expansion proves the two sum and norm identities. It preserves content: the content of a quadruple in $\mathcal{Q}_n$ divides n by its sum, while the content d of a quadruple in the other family satisfies $d^2 \mid 3n^2$, hence $d \mid n$ by lemma 2.1. In each direction a common divisor therefore divides the complemented coordinates as well. Thus the primitive subsets correspond bijectively.

Theorem 2.2 (Primitive counting bridge)**.** *For every $n \geq 1$,*

$$\mathrm{pq}(n) = \begin{cases} \frac{1}{2}F(n), & n \text{ odd}, \\ \frac{1}{2}F(n/2), & n \equiv 2 \pmod 4, \\ 0, & 4 \mid n. \end{cases} \tag{4}$$

Proof. Suppose first that n is odd. A square is $0, 1$, or 4 modulo eight, and three squares summing to $3n^2 \equiv 3 \pmod 8$ must all be one modulo eight. Thus x, y, z are odd. The four numerators in (3) are congruent modulo four, so integrality is equivalent to

$$n + x + y + z \equiv 0 \pmod 4.$$

Pair a triple with its negative. If $S = x + y + z$, then n, S are odd, and $n + S$ and $n - S$ are even residues whose sum is $2 \pmod 4$; exactly one is zero modulo four. No triple is its own negative because its norm is positive. Hence exactly one member of every sign pair gives an integer quadruple.

Primitivity is preserved for each integral inverse. A common divisor of the quadruple divides x, y, z. Conversely, the gcd d of x, y, z is odd and satisfies $d^2 \mid 3n^2$, so $d \mid n$ by lemma 2.1. Since four is invertible modulo d, (3) implies that d divides all four coordinates. This proves the first line of (4).

Now let $n = 2m$ with m odd. From $x^2 + y^2 + z^2 = 12m^2 \equiv 0 \pmod 4$, all three coordinates are even: the number of odd coordinates would otherwise be one, two, or three, none of which gives zero modulo four. Write $x = 2X, y = 2Y, z = 2Z$. Then $X^2 + Y^2 + Z^2 = 3m^2$, and X, Y, Z are odd by the preceding modulo-eight argument. The inverse formulas become

$$\begin{aligned} h &= (m + X + Y + Z)/2, \quad i = (m + X - Y - Z)/2, \\ j &= (m - X + Y - Z)/2, \quad k = (m - X - Y + Z)/2. \end{aligned} \tag{5}$$

They are integers, and any two coordinates differ by a sum of two odd numbers, so the four coordinates have the same parity. Negating the triple gives another quadruple whose corresponding coordinates sum with the first to m. Since m is odd, one of the two quadruples is entirely odd and the other entirely even.

If (X, Y, Z) is primitive, the odd quadruple is primitive: an odd prime dividing all four coordinates would divide X, Y, Z, and two does not divide the odd coordinates. The even quadruple is not primitive. If the normalized triple has gcd $d > 1$, then d is odd, $d \mid m$ by lemma 2.1, and (5), using that two is invertible modulo d, makes both quadruples divisible by d. Thus each primitive sign pair of normalized triples contributes exactly one primitive quadruple. This proves the second line. In particular, the count is $F(m)/2$, not a count of primitive *unnormalized* triples of norm $12m^2$.

Finally, if $4 \mid n$, the number of odd coordinates of a quadruple is even because their sum is even. Exactly two odd coordinates are impossible modulo four, since their squares would sum to $2 \pmod 4$. Four odd coordinates are impossible modulo eight, since their squares would sum to $4 \pmod 8$ whereas $n^2 \equiv 0 \pmod 8$. All coordinates are therefore even, so no quadruple is primitive. □

3 An explicit lattice proof of the primitive product

Write $Q(x) = x \cdot x$ on $\mathbb{R}^3$. A Euclidean lattice is called integral when all its inner products are integers; its covolume is the volume of a fundamental parallelepiped. A vector of a lattice is primitive when it is not in qL for any rational prime q.

Lemma 3.1 (Small unimodular lattices)**.** *Every integral positive definite Euclidean lattice of rank $1 \le r \le 3$ and covolume one has an orthonormal $\mathbb{Z}$-basis.*

Proof. For $r = 3$, the open ball of radius $4/3$ has volume $256\pi/81 > 8$. Minkowski's theorem [3] gives a nonzero lattice vector of squared length less than $16/9 < 2$. Its squared length is a positive integer, so it equals one. The analogous ball in rank two has area $16\pi/9 > 4$, and in rank one has length $8/3 > 2$, so the same reasoning applies in those ranks.

If $u \in L$ has squared length one, integrality implies $x \cdot u \in \mathbb{Z}$ for every $x \in L$. Therefore

$$L = \mathbb{Z}u \perp L', \qquad L' = L \cap u^{\perp},$$

because $x - (x \cdot u)u \in L'$. Orthogonal splitting preserves covolume when the first factor has length one; thus L' is integral of covolume one. Induction on rank completes the proof. □

Lemma 3.2 (Isotropic lines over an odd prime field)**.** *Let p be an odd prime and let $\mathcal{C}$ be the set of isotropic one-dimensional subspaces of $(\mathbb{F}_p^3, Q)$. Then $|\mathcal{C}| = p + 1$. For nonzero $\bar{x} \in \mathbb{F}_p^3$, the number of $\ell \in \mathcal{C}$ with $\ell \subseteq \bar{x}^{\perp}$ is*

$$1 + \left(\frac{-Q(\bar{x})}{p}\right).$$

For $\bar{x} = 0$, that number is $p + 1$.

Proof. There is a nonzero isotropic vector: the sets of squares and of numbers $-1 - y^2$ each have $(p+1)/2$ elements and therefore intersect. This gives a vector $(x, y, 1)$ of norm zero.

Fix a nonzero isotropic vector u. Nondegeneracy gives v with $u \cdot v = 1$, and $w = v - Q(v)u/2$ then satisfies $Q(w) = 0$ and $u \cdot w = 1$. The span of u, w is nondegenerate; its one-dimensional orthogonal complement has a generator e with $Q(e) = c \neq 0$. Thus the form in this basis is $2ab + ct^2$. Replacing a by $a' = -2a/c$ turns its zero equation into $a'b = t^2$. This conic is parametrized bijectively by $[u : v] \mapsto [u^2 : v^2 : uv]$. Explicitly, when its first coordinate is nonzero, rescale to $(1, s^2, s)$; its only point with first coordinate zero is $[0 : 1 : 0]$. It therefore has $p + 1$ points.

If $Q(\bar{x}) = 0$, use the same hyperbolic basis with $u = \bar{x}$. Its orthogonal plane is spanned by u, e, with norm ct^2 on $au + te$. Thus its sole isotropic line is $\mathbb{F}_p\bar{x}$.

If $Q(\bar{x}) \neq 0$, the orthogonal plane is nondegenerate. Its Gram determinant has square class $Q(\bar{x})^{-1}$: adjoining $\bar{x}$ to any basis of the plane gives Gram determinant $Q(\bar{x})$ times the plane determinant, whereas change of basis in the standard ternary space changes determinant by a square. A nondegenerate binary form has two isotropic lines when the negative of its determinant is a square, and no isotropic lines otherwise. To see this explicitly, diagonalize it as $As^2 + Bt^2$, with $A, B \neq 0$ (characteristic is odd). An isotropic line has $t \neq 0$, and s/t must be a root of $X^2 = -B/A$. There are two or zero roots according as the negative determinant $-AB$ is a square or a nonsquare. The square classes of $-Q(\bar{x})^{-1}$ and $-Q(\bar{x})$ coincide. The asserted expression follows in both cases. □

Lemma 3.3 (The neighbors of $\mathbb{Z}^3$)**.** *Put $L = \mathbb{Z}^3$. For each $\ell \in \mathcal{C}$ there is an integral covolume-one lattice L_ℓ satisfying*

$$L \cap L_\ell = M_\ell := \{x \in L : x \bmod p \in \ell^{\perp}\},$$
$$[L : M_\ell] = [L_\ell : M_\ell] = p,$$
$$pL \subseteq M_\ell, \qquad pL_\ell \subseteq M_\ell \subseteq L.$$

If $v \in L \setminus pL$ and $p^2 \mid Q(v)$, then v/p belongs to exactly one of these lattices, namely $L_{\mathbb{F}_p(v \bmod p)}$.

Proof. Choose a lift $u \in L$ of a nonzero vector of ℓ. It can be chosen with $p^2 \mid Q(u)$. Indeed, starting with any lift, replacing it by $u + pz$ changes $Q(u)/p$ modulo p by $2u \cdot z$. The linear functional $z \mapsto u \cdot z \pmod p$ is nonzero, hence surjective, so a correction z with the required value exists.

Set

$$L_\ell = M_\ell + \mathbb{Z}\frac{u}{p}.$$

All inner products are integral, since $u \cdot M_\ell \subseteq p\mathbb{Z}$ and $Q(u) \in p^2\mathbb{Z}$. The functional defining M_ℓ is surjective modulo p, so its index in L is p. The class of u/p modulo M_ℓ has order p, since $u \notin pL$ and $u \in M_\ell$. Consequently L_ℓ has covolume one. If $m + tu/p \in L$, then $p \mid t$, proving the intersection assertion. All other inclusions follow from the definition. Each L_ℓ is isometric as a lattice to $\mathbb{Z}^3$ by the preceding small-lattice lemma.

For the last assertion, membership forces the line of v mod p to equal ℓ. Conversely write $v = tu + pz$ with $p \nmid t$. The divisibility of $Q(v)$ and $Q(u)$ by p^2 gives $2t(u \cdot z) \equiv 0 \pmod p$. Thus $z \in M_\ell$ and $v/p = tu/p + z \in L_\ell$. □

The construction is independent of the normalized lift: if another lift is $tu + pz$ with $t \not\equiv 0 \pmod p$ and norm divisible by p^2, the calculation in the last paragraph shows $z \in M_\ell$; the two generators then generate the same cyclic extension of M_ℓ.

Primitivity away from p. In the double count below, passage between L and L_ℓ preserves primitivity at every prime $q \neq p$. This can be checked without using localization. A common vector $x \in M_\ell$ is in qL if and only if it is in qL_ℓ. Indeed, if $x = qz$ with $z \in L$, reduction modulo p shows $z \in M_\ell$. If instead $z \in L_\ell$, its class in the group L_ℓ/M_ℓ of order p is annihilated by q and hence is zero; again $z \in M_\ell$. For an outside vector $y = v/p \in L_\ell \setminus L$, the corresponding statement is $y \in qL_\ell$ if and only if $v \in qL$. In the latter direction, write $v = qz$. Since $v \in M_\ell$ and q is invertible modulo p, $z \in M_\ell$. Choose integers a, b with $aq + bp = 1$; then $z/p = ay + bz \in L_\ell$. The converse follows by multiplying by p, since $pL_\ell \subseteq L$.

Recall that $F(m)$ is the number of primitive vectors of norm $3m^2$ in L. These sets are finite by coordinate bounds. All the lattices L_ℓ have exactly $F(m)$ such vectors, because they are isometric to L by a bijective $\mathbb{Z}$-linear isometry. Let $\chi_p = (\frac{-3}{p})$.

Lemma 3.4 (Primitive double count)**.** *Let p be an odd prime and let m be positive and odd. If $p \nmid m$, then*

$$F(pm) = (p - \chi_p)F(m).$$

If $p \mid m$, then

$$F(pm) = (p+1)F(m) - \left(p - \left(\frac{-3(m/p)^2}{p}\right)\right)F(m/p).$$

Proof. Count pairs (ℓ, y) where y is primitive in L_ℓ and has norm $3m^2$. Their total number is $(p+1)F(m)$.

First count the pairs with $y \notin L$. The vector $v = py$ is in $L \setminus pL$ and has norm $3p^2m^2$. Conversely every primitive such v gives exactly one outside pair by the neighbor lemma. Primitivity away from p is preserved as just proved. An outside y is automatically primitive at p, because $y \in pL_\ell$ would imply $v \in p^2L_\ell \subseteq pL$. Thus the outside pairs number exactly $F(pm)$.

Suppose first that $p \nmid m$. The integer $3m^2$ is not divisible by p^2, including at $p = 3$. Therefore every vector of this norm in an integral lattice is primitive at p. Inside pairs are

precisely primitive $x \in L$ with $\ell \subseteq (x \bmod p)^\perp$. The finite-field lemma gives $1+\chi_p$ choices of ℓ for each such x: for $p \neq 3$ the factor m^2 is a nonzero square, and for $p = 3$ both Legendre symbols are zero. Therefore $(p+1)F(m) = F(pm) + (1+\chi_p)F(m)$, giving the first formula.

Now suppose $p \mid m$. An inside vector that is primitive away from p has the unique form $x = p^a z$ with $a \geq 0$ and z primitive in L. If $a = 0$, then $x \bmod p$ is nonzero and isotropic. Its unique orthogonal isotropic line is its own line; for this neighbor $x/p \in L_\ell$ by the last part of the neighbor lemma. Hence x is not primitive in that neighbor and contributes nothing. If $a \geq 2$, then $x/p \in pL \subseteq L_\ell$ for every neighbor, so again it contributes nothing.

Only $a = 1$ contributes. Write $x = pz$, with z primitive of norm $3(m/p)^2$. The vector x belongs to all $p+1$ neighbors. It fails to be primitive at p precisely when $z \in L_\ell$, or equivalently when $\ell \subseteq (z \bmod p)^\perp$. The finite-field lemma shows that the number of remaining choices is $p - (\frac{-3(m/p)^2}{p})$. Counting these pairs yields the second formula. □

Theorem 3.5 (Primitive three-square product)**.** *For every positive odd m,*

$$F(m) = 8m \prod_{p \mid m} \left(1 - \frac{\chi_p}{p}\right).$$

Proof. First $F(1) = 8$, since the vectors of norm three are precisely $(\pm 1, \pm 1, \pm 1)$.

Fix an odd prime p and a positive odd b with $p \nmid b$. The first double-counting identity gives $F(pb) = (p - \chi_p)F(b)$. Apply the second identity at $m = pb$. Its Legendre symbol is χ_p, so

$$F(p^2 b) = (p+1)(p-\chi_p)F(b) - (p-\chi_p)F(b) = p(p-\chi_p)F(b).$$

For exponent $a \geq 2$, the same identity at $m = p^a b$ has zero Legendre symbol, since $p \mid p^{a-1}b$. Induction therefore gives

$$F(p^{a+1}b) = (p+1)F(p^a b) - pF(p^{a-1}b) = p^a(p-\chi_p)F(b).$$

Removing the prime-power factors of m one at a time now gives

$$F(m) = 8 \prod_{p^a \| m} p^{a-1}(p-\chi_p) = 8m \prod_{p \mid m}(1 - \chi_p/p).$$

The local symbol agrees with the mod-six character by quadratic reciprocity, as proved explicitly in Section 4. □

The prime three. No use has been made of $p \nmid 3$ in the neighbor argument. The underlying Gram matrix of L is the identity and stays nondegenerate modulo every odd prime, including three. If $3 \nmid m$, the norm $3m^2$ has valuation exactly one, so inside vectors are still primitive at three and the isotropic-line count is one. This gives $F(3m) = 3F(m)$. If $3 \mid m$, the symbol in the second recurrence is always zero; it gives $F(3m) = 4F(m) - 3F(m/3)$, whose induction again yields $3F(m)$. Thus the prime three contributes its full power through m, while its Euler factor is one.

4 Identification of the local character

Lemma 4.1. *For every odd prime p,*

$$\chi_p = \left(\frac{-3}{p}\right) = \begin{cases} 0, & p = 3, \\ 1, & p \equiv 1 \pmod 6, \\ -1, & p \equiv 5 \pmod 6. \end{cases}$$

Proof. The value at three is zero by definition. If $p \neq 3$ is odd, quadratic reciprocity and its supplementary law [4] give explicitly

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}, \qquad \left(\frac{3}{p}\right)\left(\frac{p}{3}\right) = (-1)^{(p-1)/2}.$$

Since $\left(\frac{p}{3}\right) \in \{1, -1\}$, multiplication yields $\left(\frac{-3}{p}\right) = \left(\frac{p}{3}\right)$. The nonzero squares modulo three consist only of one. Thus this symbol is one when $p \equiv 1 \pmod 3$ and minus one when $p \equiv 2 \pmod 3$. An odd prime different from three is 1 or 5 modulo six, giving the displayed cases. □

5 Divisibility, multiplicativity, and prime-power formulas

Theorem 5.1 (The primitive arithmetic function)**.** *Write $n = 2^\alpha m$, with m odd. Then* $\mathrm{pq}(n)$ *is divisible by four, and the integer $P(n) = \mathrm{pq}(n)/4$ is given by*

$$P(n) = \begin{cases} \displaystyle\prod_{p^a \| m} p^{a-1}(p - \chi_p), & \alpha = 0 \text{ or } 1, \\ 0, & \alpha \geq 2. \end{cases} \tag{6}$$

In particular $P(1) = 1$, P is multiplicative, and, for $a \geq 1$,

$$\begin{aligned} P(2) &= 1, & P(2^a) &= 0 & (a \geq 2), \\ P(3^a) &= 3^a, & & & \\ P(p^a) &= p^{a-1}(p-1) & (p &\equiv 1 \pmod 6), \\ P(p^a) &= p^{a-1}(p+1) & (p &\equiv 5 \pmod 6). \end{aligned} \tag{7}$$

Proof. The product theorem in theorem 3.5 can be written

$$F(m) = 8 \prod_{p^a \| m} p^{a-1}(p - \chi_p).$$

Its right side is eight times an integer, including for $m = 1$, where the product is empty. The bridge (4) therefore proves divisibility of $\mathrm{pq}(n)$ by four and gives (6). Coprime positive integers have disjoint prime factors. Their odd parts therefore give a product of disjoint local factors, and their powers of two multiply according to the local function with values $1, 1, 0, 0, \ldots$. At most one of two coprime integers is even, so this local function is multiplicative as well. This proves multiplicativity of P, with $P(1) = 1$. The local formulas follow from lemma 4.1, including $p = 3$ where $\chi_3 = 0$. □

Lemma 5.2 (Content decomposition)**.** *For every $n \geq 1$,*

$$\mathrm{tq}(n) = \sum_{d \mid n} \mathrm{pq}(d). \tag{8}$$

Proof. A quadruple for positive n is not the zero quadruple, so its content c is positive. Since c divides the sum of its coordinates, it divides n. Dividing all four coordinates by c gives a primitive quadruple for n/c. Conversely, scaling a primitive quadruple for a divisor $d \mid n$ by n/d gives a quadruple of content exactly n/d. These constructions are mutually inverse, and distinct contents give disjoint sets. Counting proves the formula. □

Theorem 5.3 (The total arithmetic function)**.** *For every $n \geq 1$, the raw count* $\mathrm{tq}(n)$ *is divisible by four, and*

$$T(n) = \sum_{d \mid n} P(d). \tag{9}$$

The function T is multiplicative and, for every prime p and $a \geq 1$,

$$\begin{aligned} &T(2^a) = 2, \\ &T(3^a) = (3^{a+1} - 1)/2, \\ &T(p^a) = p^a \qquad\qquad (p \equiv 1 \pmod 6), \\ &T(p^a) = p^a + 2\frac{p^a - 1}{p - 1} \quad (p \equiv 5 \pmod 6). \end{aligned} \tag{10}$$

Proof. Every summand in (8) is divisible by four by theorem 5.1. Hence the raw total count is divisible by four, and division of this finite sum proves (9) as an identity of integers. There is no interchange of a rounded division with a sum.

If a, b are coprime, every divisor of ab factors uniquely as $d_1 d_2$, with $d_1 \mid a$ and $d_2 \mid b$. Multiplicativity of P gives

$$T(ab) = \sum_{d_1 \mid a} \sum_{d_2 \mid b} P(d_1 d_2) = \Big(\sum_{d_1 \mid a} P(d_1)\Big)\Big(\sum_{d_2 \mid b} P(d_2)\Big) = T(a)T(b).$$

Also $T(1) = P(1) = 1$. For a prime power, $T(p^a) = \sum_{j=0}^{a} P(p^j)$. At two the only nonzero summands are $P(1) = P(2) = 1$. At three the sum is $1+3+\cdots+3^a$. If $p \equiv 1 \pmod 6$, the sum is $1 + (p-1)\sum_{j=0}^{a-1} p^j = p^a$; if $p \equiv 5 \pmod 6$, it is $1 + (p+1)\sum_{j=0}^{a-1} p^j = p^a + 2(p^a - 1)/(p-1)$. These finite geometric sums give (10). □

6 The equality cases

Theorem 6.1 (Exact classification)**.** *For every positive integer n,*

$$T(n) = n \iff n = 2^{\varepsilon} m, \quad \varepsilon \in \{0, 1\}, \quad m \text{ odd}, \quad p \mid m \Longrightarrow p \equiv 1 \pmod 6 \tag{11}$$

where the last condition ranges over prime divisors. The empty factorization of $m = 1$ is allowed, so both $n = 1$ and $n = 2$ are equality cases.

Proof. Write $n = 2^{\alpha} m$ with m odd. First, for every odd prime p and every $a \geq 1$, the local formulas imply $T(p^a) \equiv 1 \pmod 3$. For $p = 3$ this follows from the geometric sum. For $p \equiv 1 \pmod 6$ it follows from $p^a \equiv 1 \pmod 3$. For $p \equiv 5 \pmod 6$, write

$$T(p^a) = 1 + (p+1)(1 + p + \cdots + p^{a-1});$$

here $p + 1$ is divisible by three. Thus $T(m) \equiv 1 \pmod 3$ by multiplicativity. Since $T(2^{\alpha})$ is one for $\alpha = 0$ and two for $\alpha \geq 1$, $T(n)$ is never divisible by three. Consequently equality is impossible if $3 \mid m$.

Assume $3 \nmid m$. Every local factor $T(p^a)$ for $p \mid m$ is odd: it is either p^a or p^a plus an even integer. Hence $T(m)$ is odd. If $\alpha \geq 1$, then $T(n) = 2T(m)$ has exact two-adic valuation one, whereas n has valuation α. Equality therefore forces $\alpha = 1$. This rules out compensating a higher power of two by larger odd-prime factors.

For $\alpha = 0$ or 1, equality is equivalent to $T(m) = m$. By (10), every local ratio $T(p^a)/p^a$ for $p \mid m$ is at least one, with equality exactly when $p \equiv 1 \pmod 6$; the ratio is strictly larger for $p \equiv 5 \pmod 6$. The product of these positive ratios equals one precisely when every ratio does. This proves necessity and sufficiency in (11), including $m = 1$. □

7 A finite sieve proving density zero

Let $\mathcal{B}$ be the primes congruent to five modulo six. Call a positive integer q *admissible* when every prime divisor of q belongs to $\mathcal{B}$. Such a q is coprime to six. Define

$$a(q) = \frac{\varphi(q)}{q}, \qquad b(q) = \frac{q}{\varphi(q)}.$$

Both are positive and $a(q)b(q) = 1$, since $q \geq 1$ gives $\varphi(q) \geq 1$.

Lemma 7.1 (Finite descent of the sieve density)**.** *For every admissible q, there is an explicitly finite admissible extension q' with $b(q') \geq b(q) + 1/6$.*

Proof. Define finite sets

$$C(q) = \{n : 1 \leq n \leq 6q,\ n \equiv 5 \pmod 6,\ \gcd(n, q) = 1\},$$
$$G(q) = \{p \in \mathcal{B} : p \leq 6q,\ p \nmid q\}.$$

There are exactly $\varphi(q)$ elements of $C(q)$. Indeed, for each invertible residue r modulo q, the Chinese remainder theorem, using $\gcd(6, q) = 1$, gives exactly one residue modulo $6q$ with residue five modulo six and residue r modulo q. Its representative in $[1, 6q]$ is positive and coprime to q; this parametrizes $C(q)$ bijectively.

Every positive integer $n \equiv 5 \pmod 6$ has a prime divisor in $\mathcal{B}$. It is divisible by neither two nor three, so all its prime factors are one or five modulo six. If all were one, their product with multiplicities would be one modulo six, a contradiction. For $n \in C(q)$, such a divisor cannot divide q and is at most $n \leq 6q$. It therefore belongs to $G(q)$.

The union bound and the exact count of positive multiples of p give

$$\varphi(q) = |C(q)| \leq \sum_{p \in G(q)} \left\lfloor \frac{6q}{p} \right\rfloor \leq 6q \sum_{p \in G(q)} \frac{1}{p}.$$

Consequently

$$\sum_{p \in G(q)} \frac{1}{p} \geq \frac{a(q)}{6}. \tag{12}$$

Set $q' = q \prod_{p \in G(q)} p$. The new primes are distinct and coprime to q, so q' is admissible and multiplicativity of the totient gives

$$b(q') = b(q) \prod_{p \in G(q)} \frac{p}{p-1}.$$

Each factor is at least $1 + 1/p$. For nonnegative numbers x_p, expanding a finite product gives $\prod(1 + x_p) \geq 1 + \sum x_p$. Using (12), we conclude

$$b(q') \geq b(q) \left(1 + \sum_{p \in G(q)} \frac{1}{p}\right) \geq b(q) \left(1 + \frac{a(q)}{6}\right) = b(q) + \frac{1}{6}.$$

All prime sets in this construction are bounded by $6q$ and finite. □

Corollary 7.2 (Arbitrarily small finite sieve products)**.** *For every real $\delta > 0$, an admissible q satisfies $\varphi(q)/q < \delta$.*

Proof. Starting with $q_0 = 1$, apply lemma 7.1 repeatedly. Each step constructs a finite positive integer, and induction gives

$$b(q_t) \geq 1 + t/6, \qquad a(q_t) \leq \frac{6}{t+6}.$$

Choose an integer $t > 6/\delta$. Then $a(q_t) < \delta$. □

Theorem 7.3 (Elementary density-zero sieve)**.** *Let E be a set of positive integers, none of which has a prime divisor congruent to five modulo six. Then*

$$\lim_{N\to\infty} \frac{|E \cap [1, N]|}{N} = 0.$$

Proof. Every member of E is coprime to every admissible q. Each complete block of q consecutive integers contains exactly $\varphi(q)$ integers coprime to q. The final incomplete block contains at most that many. Therefore

$$|E \cap [1, N]| \leq \varphi(q)\big(\lfloor N/q \rfloor + 1\big) \leq \frac{\varphi(q)}{q} N + \varphi(q).$$

Given $\epsilon > 0$, use corollary 7.2 to choose an admissible q with $\varphi(q)/q < \epsilon/2$. For every integer $N > 2\varphi(q)/\epsilon$, division of the preceding inequality by N gives

$$0 \leq \frac{|E \cap [1, N]|}{N} < \epsilon.$$

This proves the asserted limit; existence of the density was not assumed. □

Corollary 7.4 (Density of the equality set)**.** *The set $\{n \geq 1 : T(n) = n\}$ has natural density zero.*

Proof. By theorem 6.1, every such n has no prime divisor congruent to five modulo six. Apply theorem 7.3. □

In particular, the positive-asymptotic-density conjecture recorded in A354766 is false: the density exists and equals zero.

8 Verification and formalization boundary

The representation product in theorem 3.5 is proved here from the explicitly constructed neighbors, their elementary low-rank lattice classification, and a double count. The published Cooper–Hirschhorn formula agrees with it, including the prime-three factor, but is only a cross-check. The density argument is a finite construction followed by an explicit epsilon bound; reciprocal-prime divergence is not one of its assumptions.

The recovered verification program independently enumerates the original quadruples and the Hadamard inverse triples for $1 \leq n \leq 150$. It checks the corrected bridge, primitive and total counts, local product formulas, and divisibility of both raw counts by four. Those checks passed after recovery, and the resulting CSV agrees with the saved historical audit. An independent cloud computation additionally checked both defining quadruple models through $n = 300$ and primitive three-square counts at all 38 odd levels through 75, with no discrepancy. Finite computations are consistency checks, not proof steps in this paper.

The companion Lean 4.27.0 development implements the lattice classification, normalized neighbors, finite conic counts, primitive-vector double count, prime-power recurrences,

and Euler product. It also verifies the primitive A278085 complement bijection and the finite density sieve. The default dependency graph has 43 local modules with no proof admissions or added axioms. All 14 concluding dependency reports contain only `propext`, `Classical.choice`, and `Quot.sound`; neither an assumed representation formula nor native-compiler reduction is a dependency of those conclusions. The sources pin Mathlib commit `a3a10db0e9d66acbebf76c5e6a135066525ac900` and include the checks.

The mathematical derivations have undergone internal adversarial checking. This document remains a draft for independent human mathematical review. AI systems assisted exploration, checking, and drafting; that assistance is disclosed and is not itself evidence of correctness.

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